

GUANGDONG VIIYONG ELECTRONIC
TECHNOLOGY CO., LTD.

V6R0B0402C0G500NBT
(0402,C0G,6pF,±0.1pF,50V)

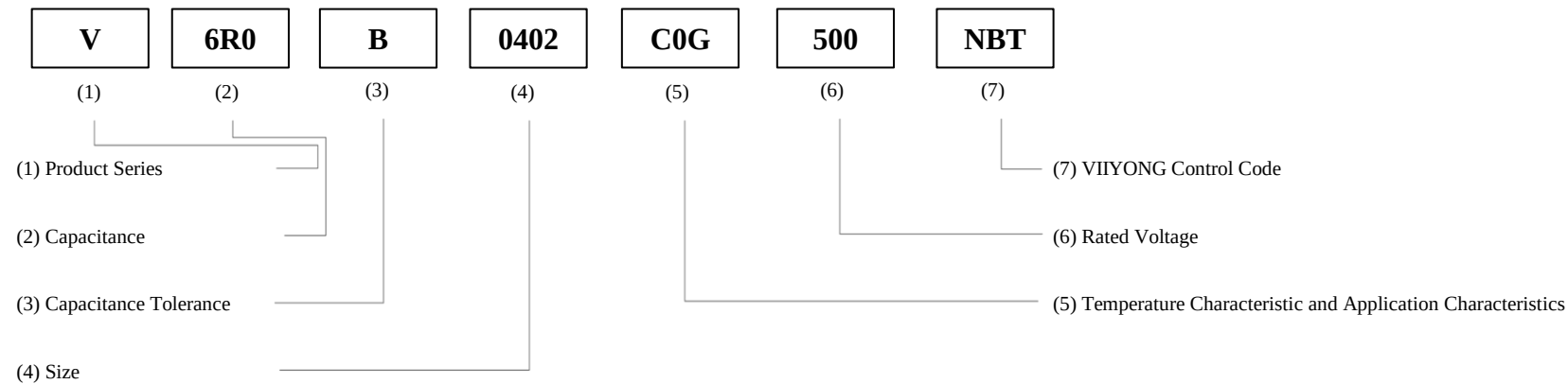
Multi-layer Ceramic Chip Capacitor Product Specification for General
Purpose

The product informations contained in this specification are as of 2025.10.09, and this specification may be revised or abolished as necessary without notice.
Please Contact your Viyyong Contact for accurate and latest Information.Should any discrepancies be identified, please verify with VIIYONG Sales.

1.Scope

This specification is applied to Multi Layer Ceramic Capacitor used for General Electronic equipment.

2.Part Number System



Rated Value (Table1)										
(4)Dimension (mm)				(2) Capacitance	(3) Capacitance Tolerance	(6) Rated Voltage	(5) Temperature Characteristic and Application Characteristics			
Length/L	Width/W	Thickness/T	L1/L2				Temp. coeff. or Cap. Change	Operating Temp. Range	Ref. Temp.	Application Characteristics
1.00±0.05	0.50±0.05	0.50±0.05	0.15~0.35	6pF	±0.1pF	50V	0±30ppm/°C	-55°C~125°C	25°C	General Purpose

3.Structure

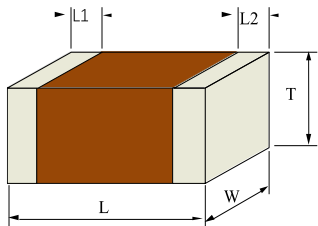


Fig.1

4.Packaging

Packaging	Quantity
7" Reel Paper Tape (W8P2)	10000 pcs./Reel

5.Specifications and Test Methods

NO.	Item	Specification	Test Method												
1	Heat treatment	Perform a heat treatment at 150+0/-5°C for 1h+/-10min and then let sit for 24+/-2hours at room temperature, then measure.													
2	Capacitance(C)	Within the specified capacitance and tolerance range.	Temperature: 18~28°C Relative Humidity: ≤80% RH Test Frequency:1MHz±10% Test Voltage: 1.0±0.2Vrms												
3	DF	DF≤1/(400+20C)													
4	Insulation Resistance(Ri)	Ri≥10000MΩ													
5	Withstanding Voltage	No breakdown or flashover during test.	Temperature: 18~28°C Relative Humidity: ≤80% RH Test Voltage: 1.0×Ur Duration: 60+/-5s												
6	Temperature Characteristics of Capacitance	≤0±30ppm/°C	Test Voltage: 3.0×Ur Duration: 60s Charge/discharge current: 50mA max.												
7	Resistance to Soldering Heat	Appearance: No defects or abnormalities. Capacitance Change:≤±0.25pF DF: Within the specified initial value. IR: Within the specified initial value.	Capacitance should be measured after sit at the temperature for 5 minutes. The reference capacitance is that of Step3 in the following table. Test voltage: ≤1.0Vrms (Refer to the individual data sheet)												
<table><tr><td>Step</td><td>Temp(°C)</td></tr><tr><td>1</td><td>25±2</td></tr><tr><td>2</td><td>-55±3</td></tr><tr><td>3</td><td>25±2</td></tr><tr><td>4</td><td>125±3</td></tr><tr><td>5</td><td>25±2</td></tr></table>				Step	Temp(°C)	1	25±2	2	-55±3	3	25±2	4	125±3	5	25±2
Step	Temp(°C)														
1	25±2														
2	-55±3														
3	25±2														
4	125±3														
5	25±2														
8	Resistance to Soldering Heat	Preheat the capacitor at 110 to 140°C for 30~60s. Immerse the capacitor in an eutectic solder solution at 260±5°C for 10±1s. The depth of immersion is 10mm. Recover it, let sit at room temperature for 24±2hrs, then measrue.													

NO.	Item	Specification	Test Method															
8	Solderability	95% min. coverage of both terminal electrodes is soldered evenly and continuously.	Immerse the test capacitor into a methanol solution containing rosin for 3 to 5s, preheat it at 80 to 140℃ for 30~60s and immerse it into molten solder of 245±5℃ for 2±0.2s. The depth of immersion is 10mm.															
9	Bond Strength of Termination	Appearance: No defects or abnormalities. Capacitance Change:≤±0.5pF	Solder the capacitor on the test substrate (shown in Fig.2), then bend the substrate at 1mm/s to 1mm (shown in Fig.3). Hold for 5±1s, then measure.															
10	Adhesion	Appearance: No defects or abnormalities.	Solder the capacitor on the test substrate (shown in Fig.2), then apply a push force F (shown in Fig.4). Hold for 10s. Adhesion:F=5N															
11	Vibration	Appearance: No defects or abnormalities. Capacitance Change:≤±0.25pF DF: Within the specified initial value. IR: Within the specified initial value.	Solder the capacitor on the test substrate, perform the test under the following conditions. Amplitude: 1.5mm Frequencies: from 10 to 55Hz, and back to 10 Hz in about 1 min. Repeat this for 2hrs each in 3 perpendicular directions X, Y, Z, total 6hrs.															
12	Rapid change of temperature	Appearance: No defects or abnormalities. Capacitance Change:≤±0.25pF DF:Within the specified initial value. IR: Within the specified initial value.	Solder the capacitor to the test substrate, follow steps 1 to 4 to do a rapid temperature change test. Cycles:5. Let sit for 24+/-4 hours at room temperature, then measure. <table><tr><td>step</td><td>Temp(℃)</td><td>Time (min)</td></tr><tr><td>1</td><td>-55+0/-3</td><td>30+/-3</td></tr><tr><td>2</td><td>25</td><td>2~5</td></tr><tr><td>3</td><td>125+3/-0</td><td>30+/-3</td></tr><tr><td>4</td><td>25</td><td>2~5</td></tr></table>	step	Temp(℃)	Time (min)	1	-55+0/-3	30+/-3	2	25	2~5	3	125+3/-0	30+/-3	4	25	2~5
step	Temp(℃)	Time (min)																
1	-55+0/-3	30+/-3																
2	25	2~5																
3	125+3/-0	30+/-3																
4	25	2~5																
13	Damp heat with load	Appearance: No defects or abnormalities. Capacitance Change:≤±0.75pF DF: DF≤ 5× initial value. IR: Ri≥500MΩ	Solder the capacitor on the test substrate, perform the test under the following conditions. Test Temperature: 40±2℃ Test Humidity: RH 90~95% Test Voltage: Ur or 1000V, whichever is smaller Duration: 500hours Charge/discharge current: 50mA max. Let sit for 24+/-4 hours at room temperature, then measure.															

NO.	Item	Specification	Test Method
14	Endurance	Appearance: No defects or abnormalities. Capacitance Change: $\leq \pm 0.3\text{pF}$ DF: $\text{DF} \leq 2 \times \text{initial value}$. IR: $R_i \geq 1000\text{M}\Omega$	Solder the capacitor on the test substrate, perform the test under the following conditions. Test Temperature: 125°C Test Voltage: $1.5 \times U_r$ Duration: 1000 ± 12 hours Let sit for 24 ± 4 hours at room temperature, then measure.

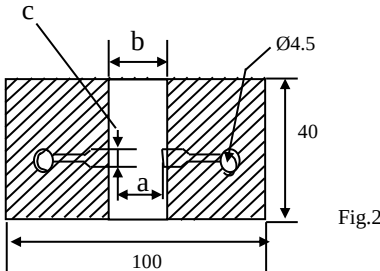
Board Flex

1. Test substrate

Material: Glass epoxy PCB

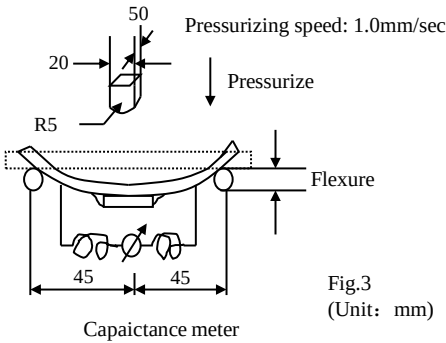
Thickness: 0.8mm (≤ 0402) or 1.6mm (≥ 0603)

2. Land Dimension

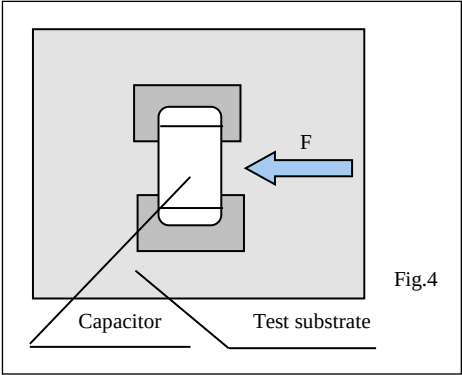


Size	Dimension (mm)		
	a	b	c
0084	0.1	0.35	0.14
0105	0.2	0.56	0.23
0201	0.3	0.9	0.3
0402	0.5	1.5	0.6
0603	0.6	2.2	0.9
0704	1.0	3.0	1.2
0805	0.8	3.0	1.3
1206	2.2	5.0	2.0
1111	2.2	5.0	2.9
1210	2.2	5.0	2.9
2220	4.4	7.5	4.8

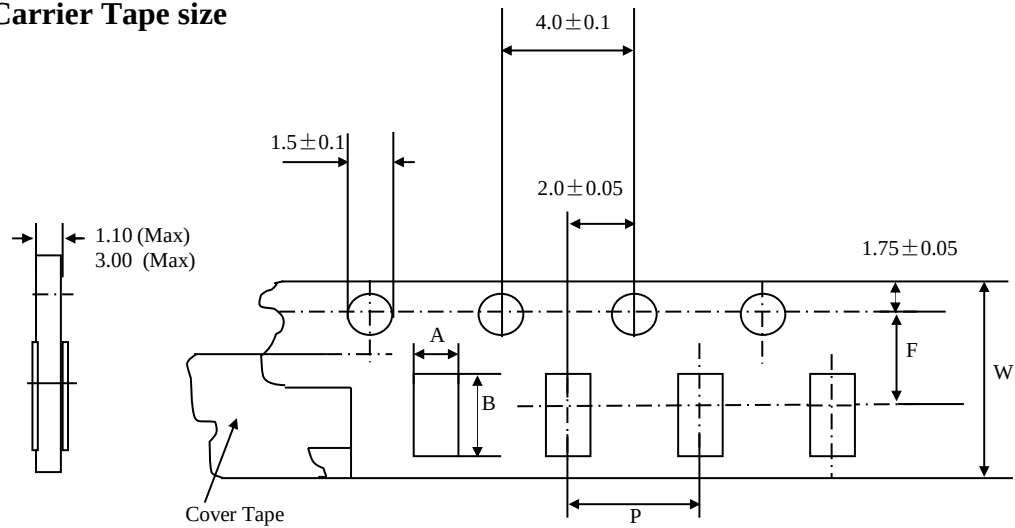
3. Pressurization Method



Terminal Strength (Adhesion)

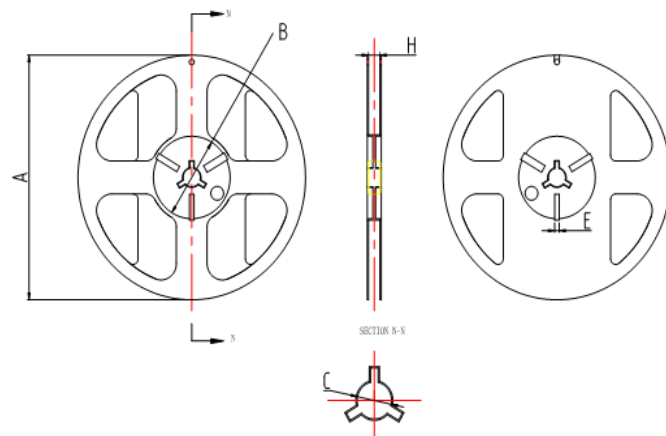


6. Carrier Tape size



Size	Dimension (mm)				
	Width of the square hole /A	Length of the square hole /B	Center distance between positioning hole and square hole /F	Square hole spacing /P	Width of carrier /W
0402	0.65(Typ.)	1.15(Typ.)	3.50±0.05	2.00±0.10	8.00±0.20

7. Disc size



Disc size	Dimension (mm)				
	A	B	C	E	H
7"	Φ178±2.0	Φ60±2.0	Φ13±1.0	4±1.0	9.5±1.0

8.Application of Technical Requirements and Precautions

8.1 Storage

8.1.1 8.1.1 Storage Period: The storage period of the product is 12 months. If it exceeds 12 months, the product shall be re-submitted for inspection.

8.1.2 Storage Conditions:

- a. Temperature: Not exceeding 35°C
- b. Relative Humidity: Not exceeding 70% RH

8.2 Environmental requirements

All products in this specification comply with the EU RoHS\REACH directive and the relevant requirements of the "VIIYONG Management Guidelines for Restriction of Hazardous Substances".

8.3 Application

8.3.1 Operating Temperature

- a. Do not use the capacitor above its upper category temperature during operation.
- b. The surface temperature and self-heating-induced temperature of the capacitor should be below its upper category temperature.

8.3.2 Operating Voltage

The operating voltage of the capacitor must be lower than its rated voltage.

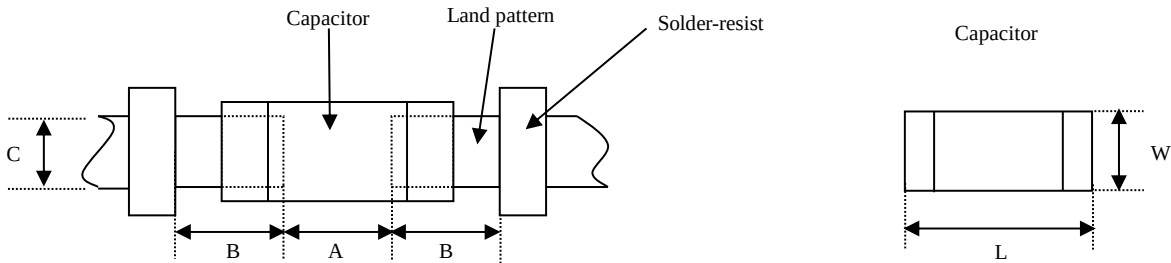
8.4 PCB Design

8.4.1. Land Pattern Design

When capacitors are mounted on a PCB, the amount of solder at the terminations has a direct impact on the capacitor performance. The greater the solder amount, the higher the stress exerted on the capacitor. Therefore, when designing land patterns, it is necessary to consider the size and structure of the land patterns (which determine solder volume). Please refer to the following designs.

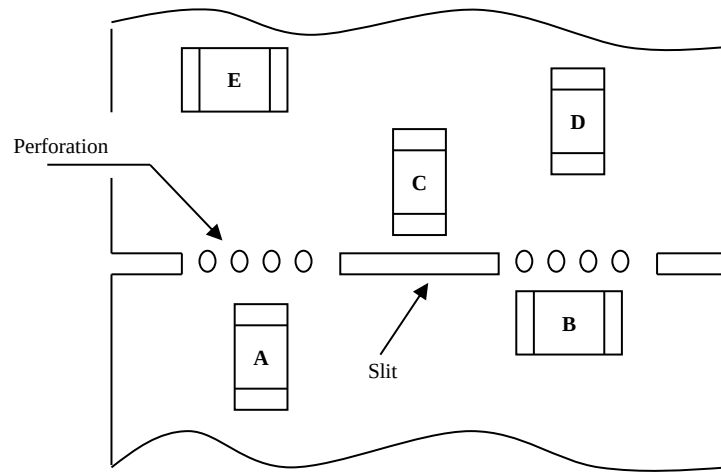
Size and recommended land dimensions for reflow-soldering are shown in the following figure and table. (Unit: mm)

Size	Length	Width	A	B	C
0402	1.00±0.05	0.50±0.05	0.30~0.50	0.35~0.45	0.40~0.60



8.4.2. Capacitor Layout on PC Board

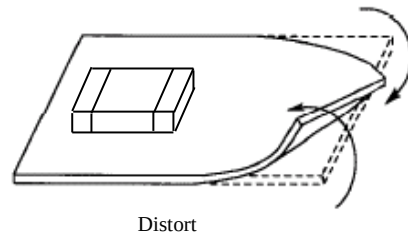
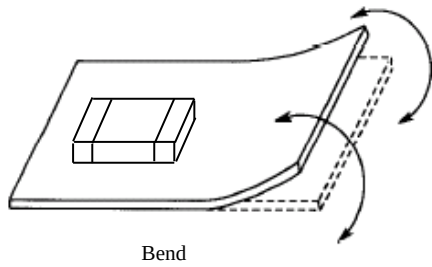
Mechanical stress varies according to the location of capacitors on PC board. The recommendation for better design is as follows:



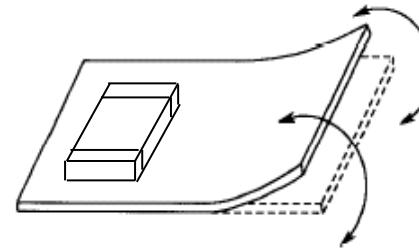
The stress in capacitors is in the following order: $A > B = C > D > E$

Pay attention not to bend or distort the PC board otherwise the capacitor may crack. Please refer to the following examples of good and bad capacitor layout.

a. Not recommended

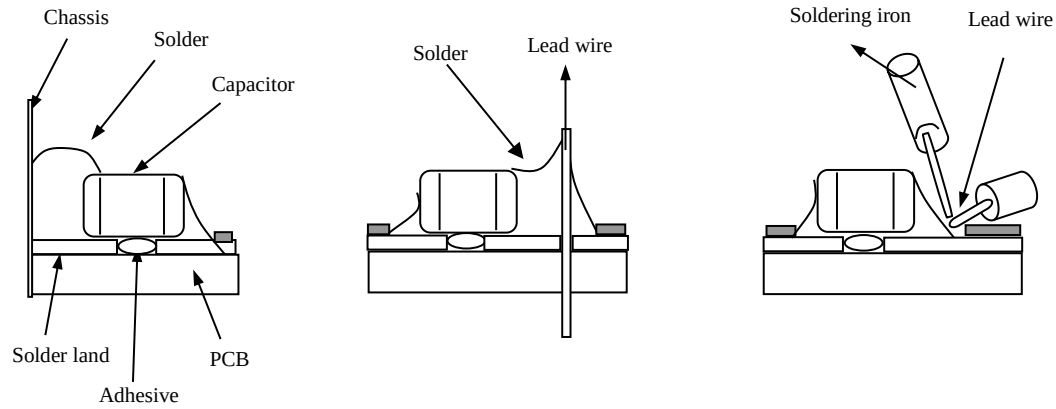


b. Recommended

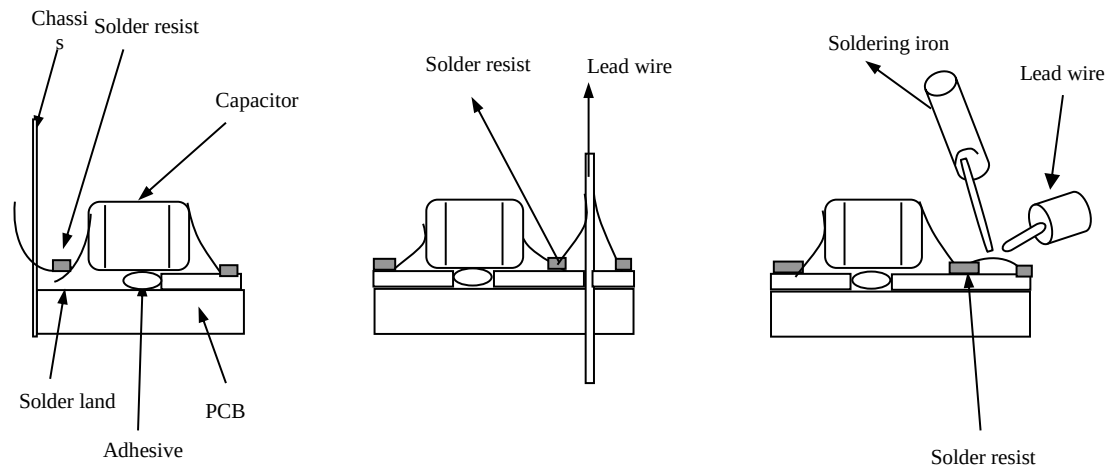


8.5 Solder Buildup and Soldering Methods

a. Examples of soldering method not recommended



b. Examples of soldering method recommended

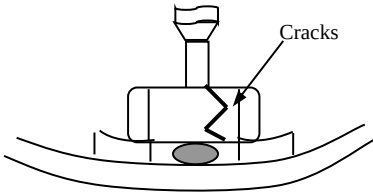
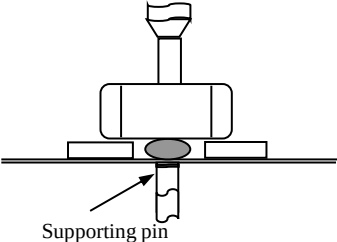
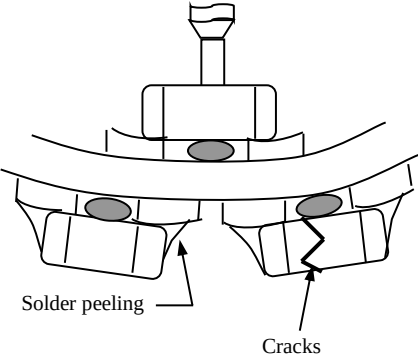
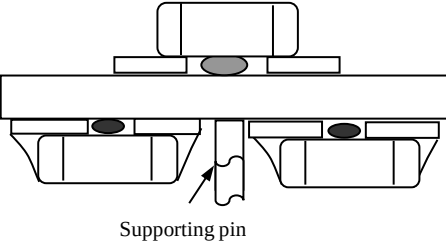


8.6 Consideration for Automatic Placement

If the mounting head is adjusted too low, it will generate excessive stress on the MLCC, leading to cracking. Please refer to the following precautions:

- a. Adjust the bottom of the mounting head to make contact with the PCB surface, without applying excessive force;
- b. Adjust the pressure of the mounting head to 1 to 3 N;
- c. To reduce the impact force from the mounting head, support should be provided from the bottom of the PCB.

Please refer to the following design examples:

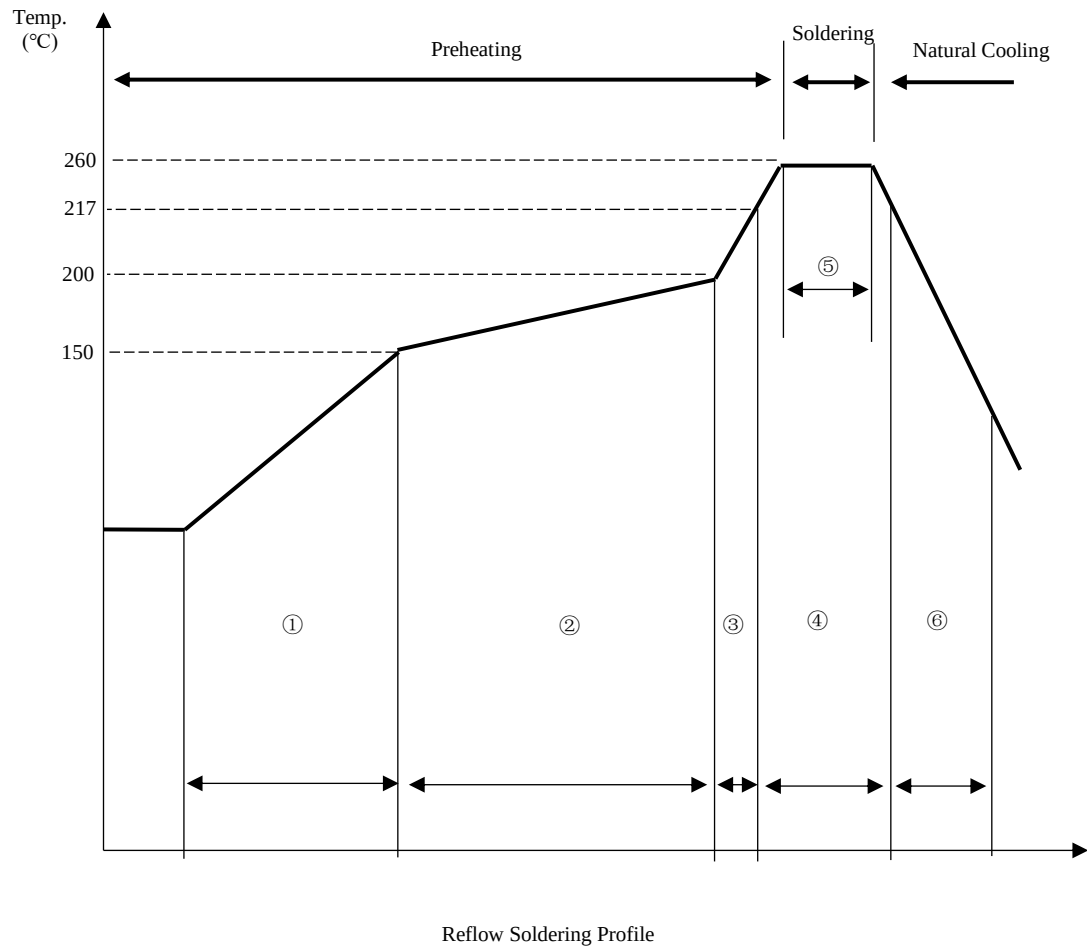
	Not recommended	Recommended
Single-sided Mounting		
Double-sided Mounting		

8.7 Soldering

8.7.1. Flux Selection

- It is recommended to use a mildly activated flux (with chlorine content less than 0.1wt%), and highly activated flux should be avoided.
- Please apply an appropriate amount of flux, and excessive flux must be avoided.
- When water-soluble flux is used, thorough cleaning is required to remove residues.

8.7.2. Design of Soldering Profile (for reference only)

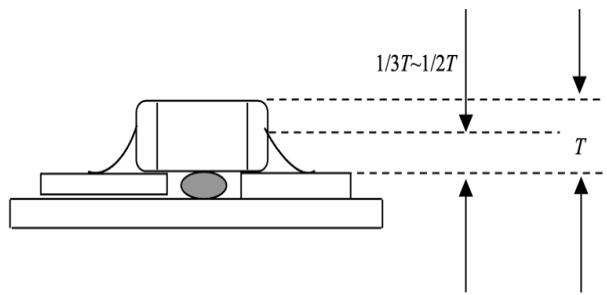


(1) Reflow Soldering Conditions

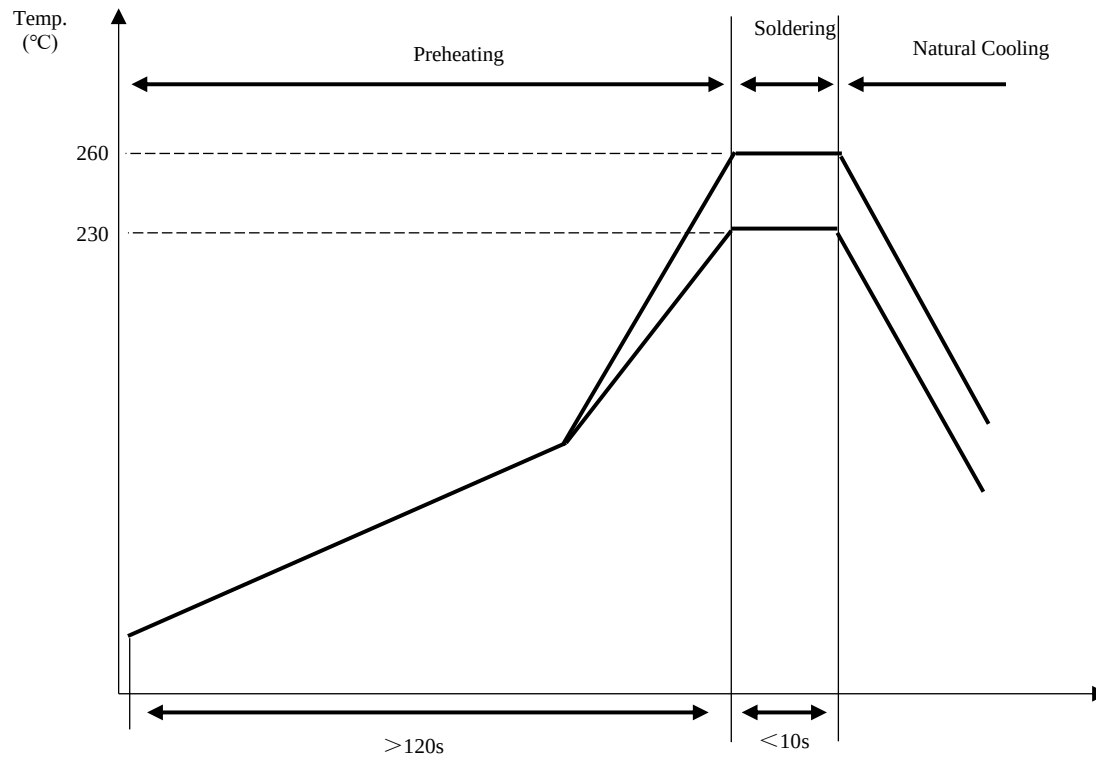
- ① Preheating 1: Heating rate: $\leq 3^{\circ}\text{C/s}$; Duration: 60s.
- ② Constant Temperature: Heating rate: $\leq 1^{\circ}\text{C/s}$; Duration at $150\text{--}200^{\circ}\text{C}$: 60–120s.
- ③ Preheating 2: Heating rate: $1\text{--}5^{\circ}\text{C/s}$.
- ④ Soldering Zone 1: Duration at 217°C : 60–150s.
- ⑤ Soldering Zone 2: Duration at $255\text{--}260^{\circ}\text{C}$: $\leq 30\text{s}$.
- ⑥ Natural Cooling: Cooling rate: $\leq 6^{\circ}\text{C/s}$.

Caution:

- a. Excessive solder will generate higher stress on the MLCC during temperature changes, leading to cracking. Insufficient solder may cause the MLCC to detach from the PCB. The ideal condition is to control the solder amount to $1/3$ to $1/2$ of the MLCC thickness, as shown in the figure below.
- b. The soldering time should be as close to the recommended time as possible; excessive time may impair solderability.
- c. If the duration for Soldering Zone 2 needs to exceed 30s, please contact VIYONG for confirmation.



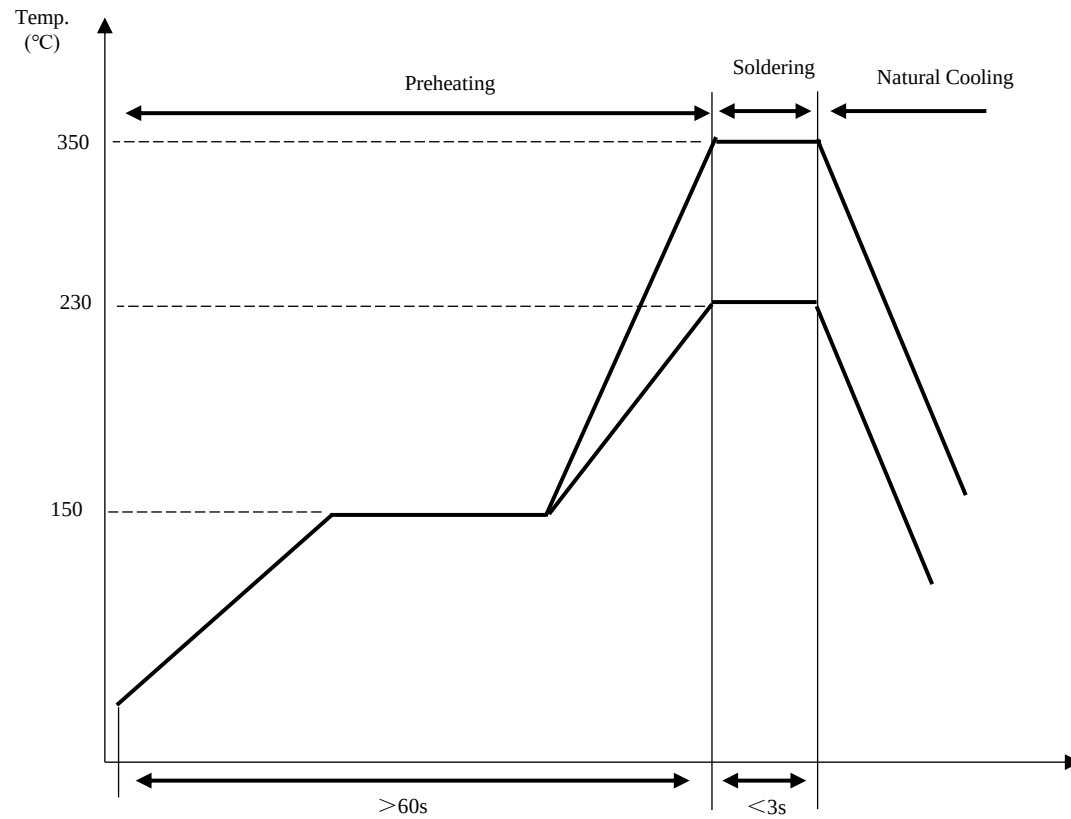
(2) Wave Soldering Condition



Caution:

- Make sure the capacitors are preheated adequately.
- The temperature difference between the MLCC and molten solder should not exceed 130°C.
- Wave soldering must not be applied to the capacitors designed exclusively for reflow soldering.

(3) Hand Soldering Condition



Caution:

- Use a 20W soldering iron with a maximum tip diameter of 1.0mm.
- The soldering iron should not directly touch the capacitor.